DM2526 – Updated Project Proposal

Chosen Project Type: [DW] Data Warehouse project with OLAP analysis

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Dataset(s) and Reference Link:
NYC TLC Trip Record Data
https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page

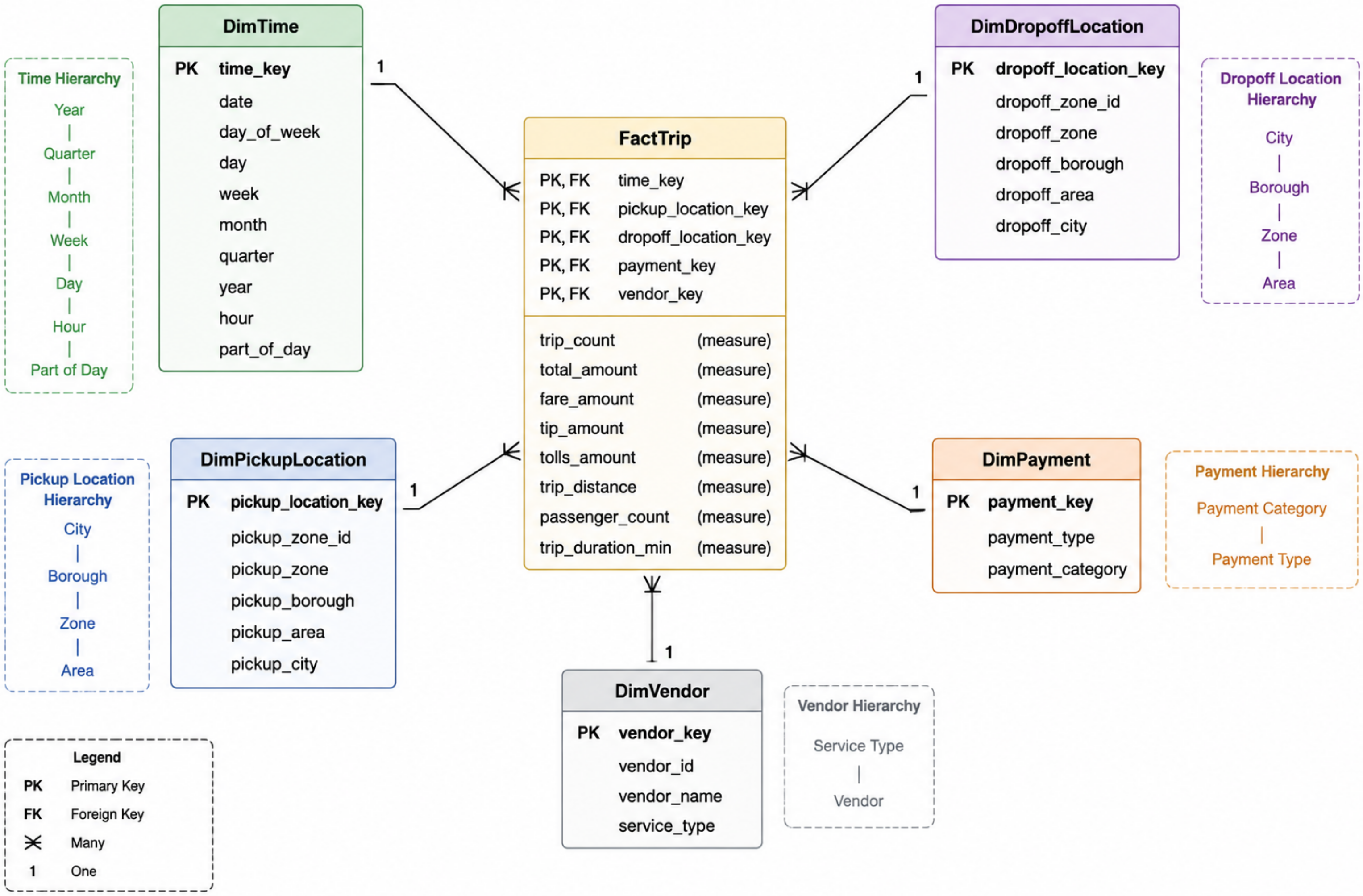
Brief Description of the Work:
The project aims to design and implement a Data Warehouse for the analysis of New York City taxi trip data. The selected dataset contains millions of records and includes temporal, geographical, payment, and trip-related information. The project will involve ETL operations for data cleaning and integration, definition of a Dimensional Fact Model (DFM), implementation of a star schema in PostgreSQL, and execution of OLAP queries to identify relevant trends and patterns in urban transportation data. The analysis will focus on ride demand, revenue trends, passenger behavior, trip duration, payment methods, and geographical distribution of taxi activity across New York City boroughs. **Technologies:**
PostgreSQL, Python (Pandas), SQL, Power BI/Tableau (optional)

Proposed DFM Structure

Fact Table: FactTrip
Measures: - fare_amount - total_amount - tip_amount - trip_distance - passenger_count - trip_duration

Dimensional Hierarchies:
1. Time Dimension: Hour → Day → Month → Quarter → Year
2. Pickup Location Dimension: Zone → Borough → City
3. Dropoff Location Dimension: Zone → Borough → City
4. Payment Dimension: Payment Type → Payment Category
5. Service Dimension: Vendor → Taxi Service Type

DFM Sketch


DimTime
PK time_key
date
day_of_week
day
week
month
quarter
year
hour
part_of_day
Time Hierarchy
Year
Quarter
Month
Week
Day
Hour
Part of Day
FactTrip
PK, FK time_key
PK, FK pickup_location_key
PK, FK dropoff_location_key
PK, FK payment_key
PK, FK vendor_key
trip_count (measure)
total_amount (measure)
fare_amount (measure)
tip_amount (measure)
tolls_amount (measure)
trip_distance (measure)
passenger_count (measure)
trip_duration_min (measure)
DimDropoffLocation
PK dropoff_location_key
dropoff_zone_id
dropoff_zone
dropoff_borough
dropoff_area
dropoff_city
Dropoff Location Hierarchy
City
Borough
Zone
Area
DimPickupLocation
PK pickup_location_key
pickup_zone_id
pickup_zone
pickup_borough
pickup_area
pickup_city
Pickup Location Hierarchy
City
Borough
Zone
Area
DimPayment
PK payment_key
payment_type
payment_category
Payment Hierarchy
Payment Category
Payment Type
DimVendor
PK vendor_key
vendor_id
vendor_name
service_type
Vendor Hierarchy
Service Type
Vendor
1
1
1
1
1
Legend
PK Primary Key
FK Foreign Key
Many
1 One